

Final report

2AMI30 – Process Intelligence in Action

1. General Information

Group number: 04

Student names and IDs:

- Carina Ungureanu (1846345)
- Gauri Sharma (2192403)
- Sujitha Navaneetha Krishnan (1771213)
- Wessel Bekx (1681621)

2. Problem Recap and Refined Bottleneck Definition

A significant number of customers abandon their shopping carts before reaching the checkout, as the perceived queueing time is too long. The management wants to understand the reasons behind this long queueing time and analyse when, where and why excessive delays are occurring. This issue is resulting in lost sales and customer dissatisfaction; hence it is very relevant for streamlined supermarket operations.

We defined a bottleneck in the supermarket checkout system as a state in which customers experience queues with a minimum length of 4 people per active counter, before they enter any queue or experience increasing waiting times.

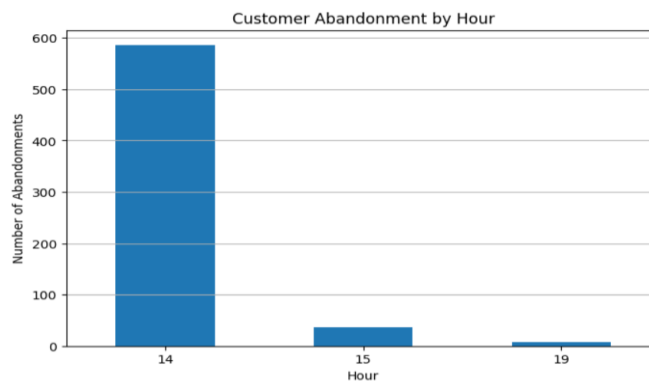


Figure 1: Number of customer abandonments by hour

Bottlenecks occur at the checkout counter, specifically during the waiting phase between a customer entering the queue and the first item being scanned. They concentrate at three distinct times during a day, namely at 14:00, 15:00 and 19:00, as it can be seen in Figure 1, where the bottleneck is reflected in the number of cart abandonments, which only happen during these times. Figure 2 shows the frequency of when these cart abandonments happen, and it can be seen that in 39 out of the 40 days analyzed, there are cart abandonments at 14:00, in 21 of the days, there are cart abandonments at 15:00 and there is one day with cart abandonments at 19:00.

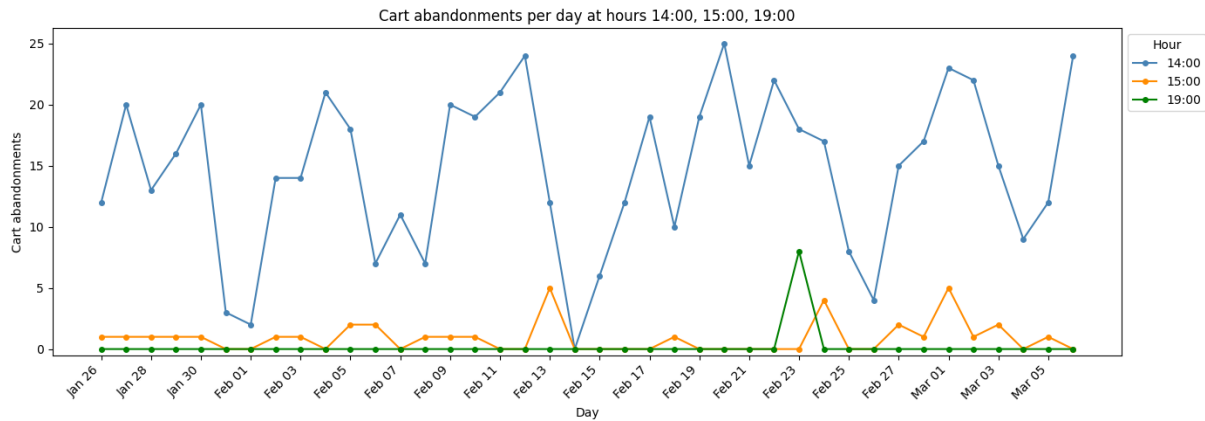


Figure 2: Cart abandonments at 14:00, 15:00 and 19:00 for all the days

3. Bottleneck Analysis and Diagnostics

To identify the main drivers of the bottleneck at the checkout, we analysed several factors affecting queue buildup, waiting times and cart abandonment. This included **queue length** and **waiting time**, the **number of open/active counters**, **shift handovers**, and **customer arrival rates** at the checkout counter. Queue length and waiting time were used as measures of bottleneck severity, while the remaining factors were investigated as possible causes.



Figure 3 : Median waiting time for days experiencing high cart abandonments

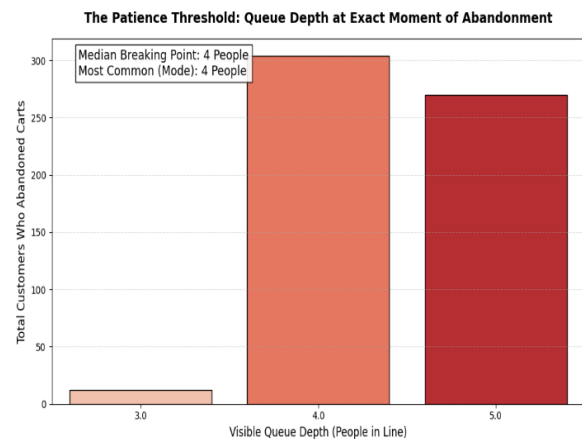


Figure 4: Queue length when abandonments occur

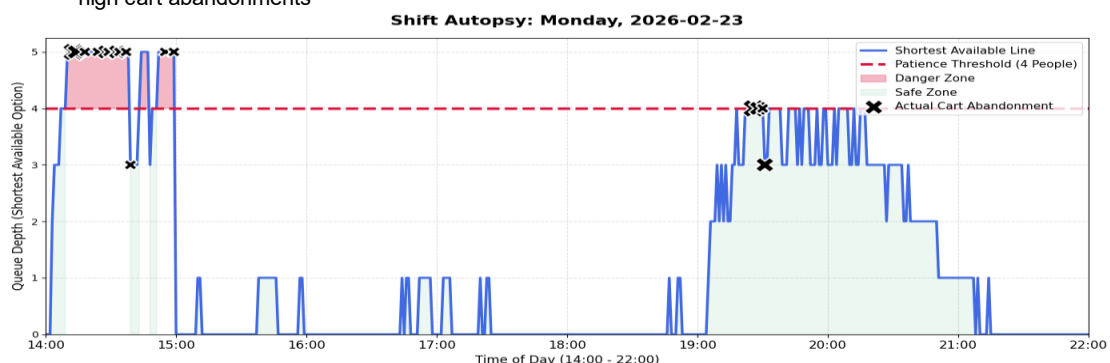


Figure 5: Relation between minimum queue length and cart abandonments

The strongest evidence of a bottleneck comes from the relationship between queue length and cart abandonment. The analysis we performed showed that the customers abandon their carts almost exclusively when the shortest available queue reaches four people as seen in Figure 5. This suggests that the customers are reacting directly to the queue conditions they

encounter when they decide whether to join a queue. As seen in Figure 4 both the median and the most common queue depth (length) when the abandonment occurs is equal to four. Additionally, we identified 14:00 as the critical hour (Figure 3), with median waiting times always around 10 minutes and most of the time below one minute during other hours. This pattern is observed on 39 out of 40 observed days; this solidifies that this phenomenon is consistent.

Queue length and waiting time can be considered as direct drivers of the bottleneck. The relationship observed is causal since customers see the shortest available queue before joining and decide based on what they see. Longer queues lead to longer waiting times, which increases the likelihood of abandonment. Additional evidence can be supported from the simulation results, that is, after increasing checkout capacity and reducing queue buildup, cart abandonment rates dropped to almost zero. Since change in queue conditions, showed change in abandonment behaviour, this solidifies the causation relationship. It is primarily a structural issue since the same queue buildup occurs repeatedly across almost all observed days, indicating that the problem is caused by the way checkout capacity is planned.

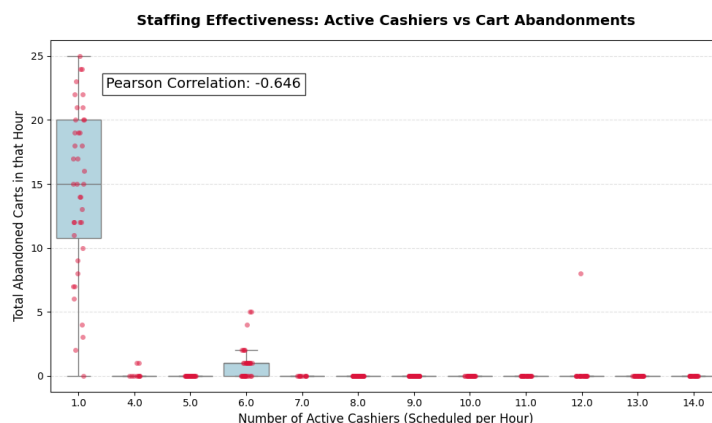


Figure 6: Correlation between active cashiers and abandonments

The **number of open counters** is strongly related to cart abandonment as seen above (Figure 6) by the Pearson correlation (-0.646) between the number of active cashiers and abandonment counts in a particular hour. It indicates a strong negative relationship, meaning that cart abandonment decreases as the number of active cashier increases. The total number of abandoned carts in that hour can reach up to 25 customers per hour when only one counter is open in contrast to almost no abandoned carts when more counters are open. Hence, this factor is also a driver of the bottleneck, since increasing checkout capacity by opening additional counters reduces queue lengths and waiting times. There is again a causal relation as seen in Section 5 about the simulation results, as adding extra counters at 14:00 eliminates the abandonment spike almost entirely. It is a structural issue since staffing and number of counters open are determined in advance based on the schedules. Hence, it is an issue in the planning.

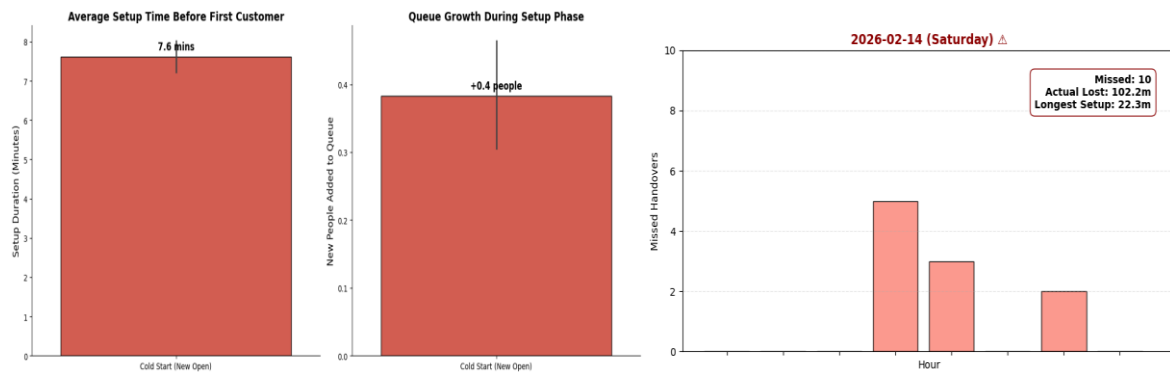


Figure 7: Queue growth observed during setup phase

Figure 8: Missed handovers for a day

It was found that **shift handovers** create a temporary bottleneck at the checkout counter. When a cashier starts shift on a closed register, an average setup period of 7.6 minutes is required before the first customer can be served. During this time, as seen in Figure 7, queues continue to grow. On the worst day as shown in Figure 8, the cumulative effect becomes significant. For instance, 10 inefficient handovers resulted in more than 102 minutes of lost checkout capacity. Therefore, it can be considered as a secondary driver of the bottleneck. The relationship appears to be causal as seen later in Figure 15 from the simulation results with the proposed intervention. The setup delay is removed by allowing cashiers to take over already open counters which reduces the temporary queue buildup. However, this would not fully explain the persistent congestion observed at 14:00. That is, even if all handovers were performed efficiently, the bottleneck caused by insufficient checkout capacity would remain. It is an operational issue, since the problem arises from how the shifts are handed over rather than from the staffing structure itself.



Figure 9: Visualising customer cashier ratio with corresponding waiting time

Initially, **customer arrival rate** was investigated as a potential driver of the bottleneck, however with further analysis, it was found that higher arrivals do not necessarily lead to congestion. That is, 14:00 has the lowest average customer volume of the day but longest waiting times, while 19:00 has the highest customer volume and considerably shorter waiting times, while 19:00 has the highest volume and considerably shorter waiting times. However, a key factor is the **customer-to-counter ratio**, which indicates how much checkout capacity is available for the number of customers in the store. At 14:00 this ratio has the highest value, leading to longer queues and substantially higher waiting times. Later in the day, the number of customers increase, but since additional counters are opened, it does not lead to much congestion as more customers can be accommodated. Hence, it is a direct driver of the bottleneck. Moreover, the relationship appears to be causal, as increasing checkout capacity in the simulation (at 14:00) reduced waiting times and eliminated the spike in abandoned carts. This is primarily a structural issue, as the customer-to-counter ratio is determined by staffing and counter scheduling decisions.

4. Proposed Intervention

To address the main structural and operational issues identified in the analysis above, we propose the following two interventions:

Intervention 1: Shifting the cashier schedule

Process data reveals a severe structural bottleneck occurring at 14:00 on 39 out of 40 observed days, where the median customer waiting time spikes past 6 minutes. This is driven by an unmanageable customer-to-counter ratio during this peak hour. Historical data confirms a "spillover effect": when additional counters are opened later at 15:00, waiting times drop significantly. However, this capacity relief currently arrives too late, forcing customers to endure the peak queue. We propose permanently opening 2 additional counters at 14:00 to lighten the load on the cashiers and reduce the queue length.

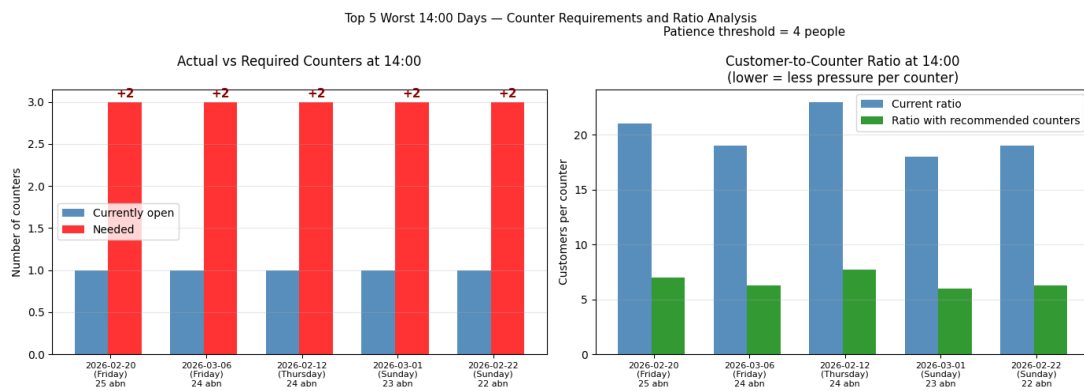


Figure 10: Adding 2 more counters (left) decreases customer counter ratio (right)

Intervention 2: Taking over already open counters

On the operational side, process logs show that when an incoming cashier starts a shift, they routinely initialize a closed register. As shown in Figure 11 this "cold start" adds an average of 7.6 minutes of setup time and in total over 13 hours of additional waiting time. During this delay, the lane remains non-functional, causing queue buildup and increased waiting times. We propose a mandatory **"Warm Takeover" protocol**: incoming cashiers must take over a counter that is already open, if possible.

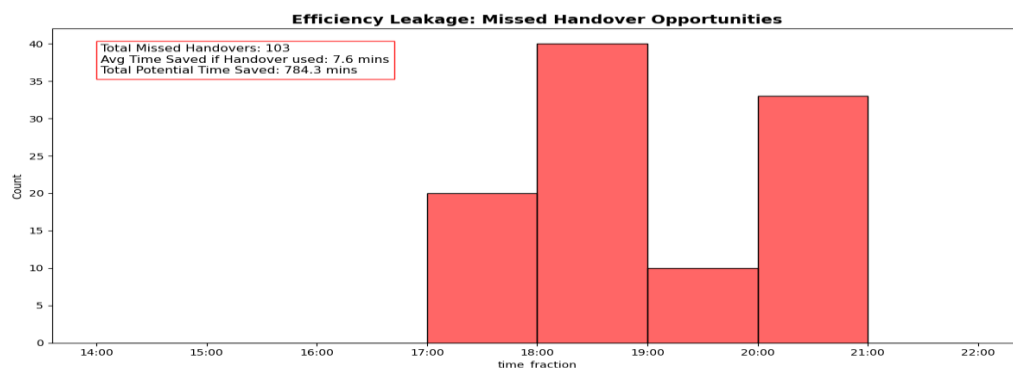


Figure 11: Visualising the cold start

5. Simulation Results and Impact Evaluation

In this section we will compare the baseline data from the checkout process against the new data with our implemented interventions. Remarkably the implementation of our interventions

reduced the total number of abandonments to zero. To determine whether this is a product of random variance or whether our interventions brought significant change in the process we perform multiple analyses below.

First, we will evaluate the macro level impact of our interventions by looking at the distribution of the queue lengths experienced by the customers immediately upon arrival at the checkout area.

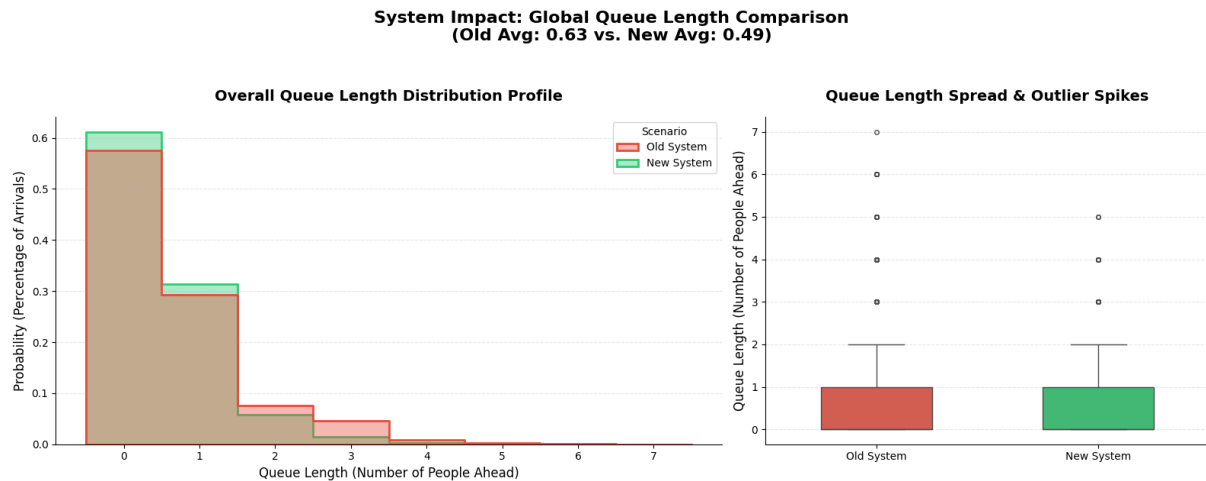


Figure 12: Queue length comparison between the old system and the new system

As illustrated above in Figure 12, our interventions structurally optimized the checkout experience. Under the new configuration, the right tail of the distribution is significantly reduced with no queue being longer than 5 people, whereas in the old configuration it could go as high as 7. We can also see that the number of queues being of length 0 or 1 has increased notably and the average queue length is reduced from 0.63 to 0.49. The Welch's t-test confirms that the structural reduction in waiting times is statistically significant ($p = 1.66 \times 10^{-31}$), which validates that the operational changes are systemic and not just random variance.

To quantify what drives the waiting times, an OLS regression analysis was compared across both the old and the new data. The models showed a structural shift, marked by a decline in the explanatory power as the R^2 value dropped from 0.557 to 0.404. This shows that in the old data, the wait times were heavily dictated by predictable factors such as handover stalls or capacity deficits. The optimized stores waiting times are more dictated by low impact stochastic noise, which reduces the model's fit.

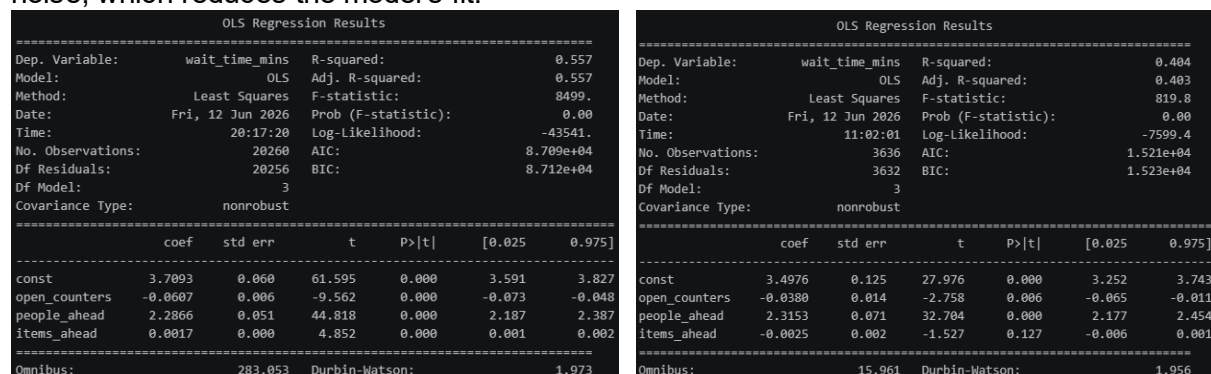


Figure 13: Regression models for waiting time of both old(left) and new(right) data

The 2 regression models in Figure 13 show that the impact of certain variables changes due to our interventions. While `items_ahead` was highly significant in the legacy system, it completely loses statistical significance in the optimized model ($p=0.127$). Meanwhile, the

number of customers in line remained constant, with **people_ahead** shifting slightly from 2.29 to 2.32 minutes per person ($p < 0.001$). This invariance proves that while human processing time is an operational constant, the intervention successfully compressed physical line lengths to prevent this 2.32-minute penalty from multiplying. Finally, systemic stabilization reduced the urgency for extra capacity, shifting the **open_counters** coefficient from -0.0607 to -0.0380 ($p = 0.006$) as severe backlogs were eliminated.

To evaluate our intervention of the mandatory "warm takeover" we compare the distributions of the waiting time for the customers in a 30-minute window after the takeover occurs. This time window is chosen to consider possible spillover effects in the waiting time after a handover. We also take into account that due to our change in cashier scheduling the number of active counters may impact this waiting time distribution, so only times when the number of active counters is the same are used for this analysis.

Pure Efficacy: Aggregated Wait Times Controlled for Staffing Capacity

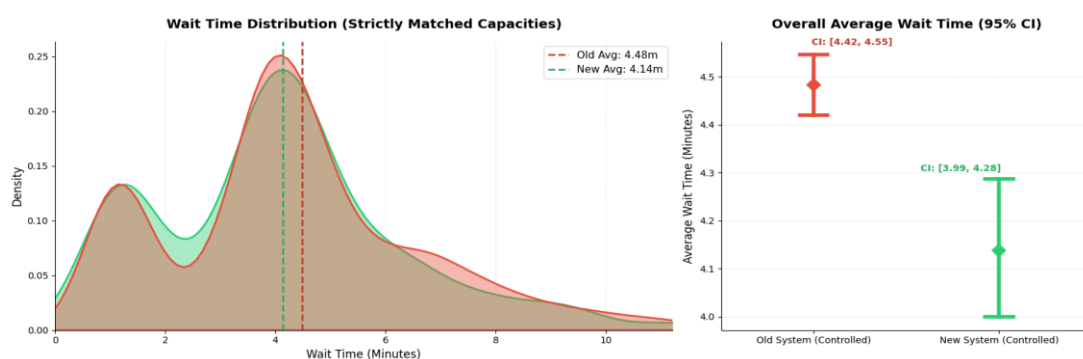


Figure 14: The distribution of waiting time after a shift hand-over has taken place controlled for staffing capacity changes

As can be seen in Figure 14, our intervention suppresses the right tail in the distribution and thus reduces the number of extreme delays due to inefficient handovers. The intervention reduced the average waiting time from 4.48 minutes to 4.12 minutes under our new policy. The 95% confidence intervals confirm that the reduction in waiting time is statistically significant with them exhibiting zero overlap. This is also shown by the p-value of $2.3543e-05$ which shows that the change in waiting times is statistically significant.

We also want to see the impact of the "warm takeover" protocol together with the scheduling change. For this we look at the same 30-min handover window, but do not control for the staffing capacity in this case so we can see how the customer experiences the waiting lines in all cases and not just in cases where the capacity is the same.

Event-Driven Impact: Isolating the Shift Handover Bottleneck

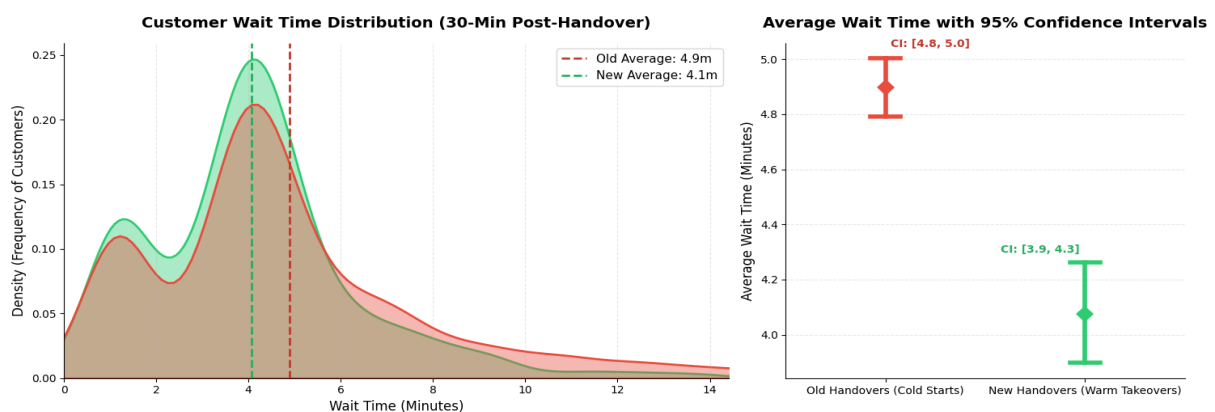


Figure 15: The distribution of waiting times after a shift-handover not controlling for staffing capacity

The distribution of the wait time in Figure 15 shows a bigger impact than it does while controlling for the staffing capacity. This is to be expected, since the scheduling changes aimed to reduce waiting times over the entire day. The right tail of the distribution gets suppressed even harder, meaning that extreme waiting times barely occur anymore. On average the waiting time gets reduced from 4.9 minutes to 4.1 minutes. The 95% confidence interval of [4.8, 5.0] of the original data and the post-intervention interval of [3.9, 4.3] exhibit absolutely no overlap, proving the change is not a product of random variance.

Lastly, we want to evaluate how the interventions influenced the customer patience and abandonment behaviour. To do this we look at the conditional probability of a customer abandoning their cart during low congestion arrival conditions: $P(\text{abandon} \mid \text{min queue} < 4)$. This metric isolated the proportion of customers who arrived when the shortest available register line was below the standard patience threshold of 4 people and still chose to abandon their carts. This queue is tracked the moment a customer arrives at the checkout, area mirroring the customer's experience of observing the shortest possible line.

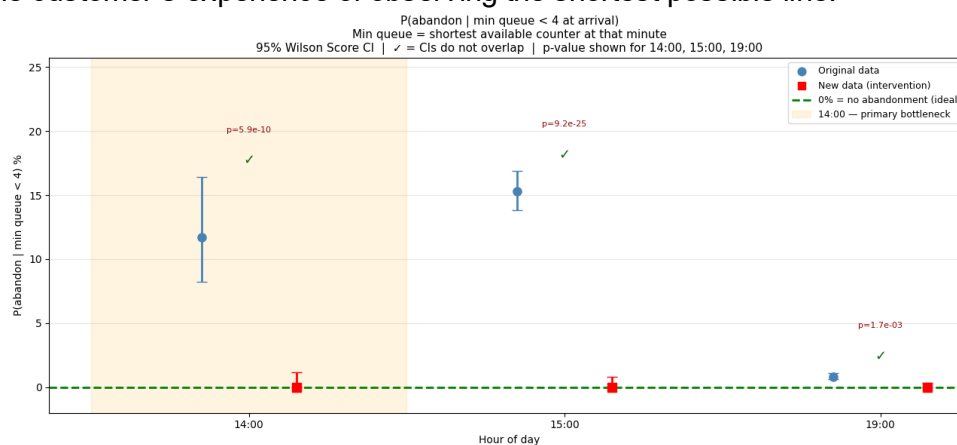


Figure 16: Cart abandonment under low congestion conditions before and after interventions

Figure 16 shows that, in the baseline simulation cart abandonments occurred at a rate of 11.7% during the 14:00 bottleneck hour, even though the physical line was below the patience threshold. The same can be seen for the other times where abandonments occurred, namely at 15:00 and 19:00. Following the implementation of the interventions, this conditional probability dropped to exactly 0% across every single operating hour of the day. The new 14:00 confidence interval shrinks tightly to [0.0%, 1.2%] and the 15:00 interval to [0.0%, 0.8%]. Because the original lower bound at 14:00 (8.2%) remains far above the post-intervention upper bound (1.2%), the complete lack of interval overlap confirms that this behavioural improvement is highly statistically significant.

6. Critical Reflection

Our main goal with suggested interventions was to decrease the cart abandonment and improve customer experience. The interventions suggested were for a cashier to take over an already open counter if possible, when a shift-handover happens and to open more counters at 14:00 which would lead to less queue buildup. These suggestions were offered because taking over an already open counter took less time as opposed to starting a fresh counter.

One assumption behind the handover intervention is that there was no formal policy requiring cashiers to open a new register instead of taking over an existing one. The event logs show

this behaviour, but not why it happened, so we assumed it could be changed through process redesign. If an existing policy or operational constraint explains the behaviour, the intervention may not be feasible. Our interpretation may also be influenced by repeatedly observing this pattern in the data.

We include price check as noise in analysis, however if it was done for an item towards the end of the scanning process for a customer may influence the waiting time of other customers at that counter. The model also assumes that cashiers are trained adequately before the first day recorded in the data because otherwise inefficiency could lead to increase in waiting time. Customer abandonment is primarily modelled as a response to congestion and resource constraints, whereas in reality, factors such as product unavailability, dissatisfaction with service, or personal time constraints may also influence abandonment decisions. The model further assumes that customers either join a queue or abandon their purchase and does not account for queue switching behaviour. Waiting time for the person in checkout area before entering a queue is assumed to be negligible, but if multiple customers are present at the same time, this assumption may underestimate their actual waiting time. In addition, the model assumes that customers always behave rationally by selecting the shortest available queue. In reality, queue choice may also be influenced by factors such as counter location, familiarity, perceived cashier speed, or nearby product displays.

The modelling choices improved process performance by reducing baseline waiting times and eliminating the statistically significant impact of basket workload on waiting time. In both (original and new log) systems, the number of customers ahead remained the dominant predictor of waiting time ($\beta \approx 2.3$, $p < 0.001$). The effect of additional open counters remained significant but weakened after intervention, suggesting reduced sensitivity to staffing levels. Most notably, the influence of items ahead became statistically insignificant after the intervention scenario ($p = 0.127$), indicating that waiting times became less dependent on basket workload. This suggests a more stable and efficient queueing process.

While the results indicate that the proposed interventions improved process performance, several limitations should be considered when interpreting the findings. The evaluation is based on a simulation model, meaning that the estimated improvements depend on the assumptions used to represent operational behaviour and may not fully reflect real-world conditions. Furthermore, certain analytical choices, such as the selection of observation windows, thresholds, and performance measures, may influence the magnitude of the reported effects, and alternative modelling choices could yield different outcomes. The large sample size also increases statistical power, making it possible for relatively small differences to appear statistically significant even when their practical impact is limited. Finally, the analysis is constrained by the information available in the event logs, which may not capture all factors influencing customer behaviour and operational performance, potentially introducing biases that affect the estimated intervention effects. Therefore, while the findings provide strong evidence that the interventions are beneficial, the precise magnitude of the improvements should be interpreted with appropriate caution.

7. Self-Reflection

This project significantly improved our understanding of process mining and data-driven decision making. One of the most valuable lessons was learning how to properly define a bottleneck rather than simply identifying where delays occur. We learned to identify the factors causing a bottleneck. Another important insight was understanding the need to analyse the system from the perspective of the decision maker.

The project also reinforced the importance of evidence-based recommendations. Exploratory Data Analysis (EDA) played a key role in this process by helping us identify patterns and focus our analysis on the most relevant aspects of the process. Rather than proposing solutions based on intuition, we learned to support every intervention with quantitative evidence and evaluate its impact through a rigorous cost-benefit mindset. We gained a deeper understanding of distributions, confidence intervals, and the distinction between statistical significance and practical significance. In several cases, we found that relying solely on averages or medians could hide important behavioural patterns, leading us to analyse distributions and temporal dynamics instead. Combining information from multiple event logs and studying the process from the perspective of different actors also highlighted the importance of understanding how system state evolves over time.

In future projects, we will place greater emphasis on understanding the problem context before beginning the analysis. One of the biggest lessons learned was that defining the right problem statement is often more important than doing a random analysis. Looking back, we believe that adopting the customer's perspective earlier in the project could have led to additional insights and potentially stronger interventions. We also learned that mentally simulating how a process unfolds in real life can reveal behavioural nuances that are not immediately visible in the data.

Our group worked very effectively throughout the project. Each member contributed according to their strengths and was willing to challenge assumptions and question proposed approaches. These discussions often led to improved analyses and prevented us from accepting conclusions without sufficient evidence. Team members supported one another when complex technical or analytical issues arose, creating a collaborative environment that encouraged both learning and problem solving.

Artificial Intelligence (AI) played an important supporting role during the project. AI was particularly useful when implementing variables and analyses that had already been carefully designed and justified by the team. In these situations, AI accelerated coding tasks and helped identify potential flaws or overlooked considerations in our reasoning. However, AI was less reliable when asked to independently generate evidence or identify process insights without sufficient context. Generic prompts often resulted in superficial conclusions that were not well aligned with the specific characteristics of the process under study.

To verify AI-generated outputs, we adopted a critical review process. Each of us critically reviewed its recommendations, and additionally also asked AI to identify weaknesses in its own recommendations and challenged its assumptions through follow-up questions. Any proposed methodology or code was reviewed against our understanding of the process before being implemented. This approach allowed us to benefit from AI as a productivity tool while maintaining confidence in the correctness of our final analysis.